

Algebra II with Trigonometry

Chapter 6.1

Olympiad Solutions

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[4506904_alg-2trig-h-chp-6-1-note-docx.pdf](#)

Solutions

1. Let t be a real number such that the terminal point determined by t on the unit circle has rational coordinates (x, y) . If t is not a multiple of $\frac{\pi}{2}$, show why both x and y must be of the form $\frac{a}{c}$ and $\frac{b}{c}$ for integers a, b, c with $a^2 + b^2 = c^2$. Give an example of such a point and determine the reference number for this t .

Solution:

On the unit circle, the terminal point for t is $(x, y) = (\cos t, \sin t)$ with $x^2 + y^2 = 1$. If both x and y are rational, then the pair (x, y) is a rational point on the unit circle.

If t is not a multiple of $\frac{\pi}{2}$, neither coordinate is exactly 0 or ± 1 . Consider $x = \frac{a}{c}, y = \frac{b}{c}$ with a, b, c integers, $c > 0$, and $\gcd(a, b, c) = 1$. Then,

$$x^2 + y^2 = \frac{a^2 + b^2}{c^2} = 1 \implies a^2 + b^2 = c^2.$$

Thus, (a, b, c) forms a Pythagorean triple. Therefore, all rational points on the unit circle arise in this way.

Example: Take the Pythagorean triple $(3, 4, 5)$. So possible points are

$$\left(\frac{3}{5}, \frac{4}{5}\right) \quad \text{or} \quad \left(\frac{4}{5}, \frac{3}{5}\right)$$

The reference number for t such that $(\cos t, \sin t) = (\frac{3}{5}, \frac{4}{5})$ is $t_{ref} = \arccos(\frac{3}{5})$.

2. The terminal point of t on the unit circle is (x, y) . Quickly determine, without a calculator, the terminal point for $5\pi - t$ and explain the symmetry involved.

Solution:

Since $5\pi = 2\pi \cdot 2 + \pi$, so $5\pi - t = \pi - t$ (modulo full revolutions the position is the same as $\pi - t$), and on the unit circle:

$$(\cos(\pi - t), \sin(\pi - t)) = (-\cos t, \sin t)$$

Therefore, the terminal point for $5\pi - t$ is $(-x, y)$.

Symmetry: This is reflection over the y -axis, as $\pi - t$ means "mirror t across the y -axis". The same holds for $5\pi - t$ due to the period of 2π .

3. Consider the sequence $t_n = \pi + (-1)^n \frac{\pi}{6}$ for $n = 1, 2, 3, \dots$. For which values of n does the terminal point determined by t_n lie on the coordinate axes? Justify your answer mentally.

Solution:

Terminal points on coordinate axes have $x = 0$ **or** $y = 0$, i.e., t_n is a multiple of $\frac{\pi}{2}$ ($0, \pi/2, \pi, 3\pi/2, \dots$).

Let's write out the sequence:

- For odd n , $t_n = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.
- For even n , $t_n = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$.

Neither $\frac{5\pi}{6}$ nor $\frac{7\pi}{6}$ are multiples of $\frac{\pi}{2}$. **Therefore, for no value of n does the terminal point lie on the axes.**

4. Estimate whether the sum $S = \cos(\frac{2\pi}{7}) + \cos(\frac{4\pi}{7}) + \cos(\frac{6\pi}{7})$ is positive, negative, or zero, and explain your reasoning using the symmetry of the unit circle.

Solution:

Observe that $\cos(\theta) + \cos(2\pi - \theta) = 2 \cos \theta$, so the cosines are symmetric about π .

Notably, $\cos(\pi - x) = -\cos x$. Let us check the numbers:

- $\frac{6\pi}{7} = \pi - \frac{\pi}{7} \implies \cos\left(\frac{6\pi}{7}\right) = -\cos\left(\frac{\pi}{7}\right)$
- $\frac{4\pi}{7} = \pi - \frac{3\pi}{7} \implies \cos\left(\frac{4\pi}{7}\right) = -\cos\left(\frac{3\pi}{7}\right)$
- $\frac{2\pi}{7}$ is unique.

But the sum S is well-known in trigonometric circles:

$$\sum_{k=1}^{n-1} \cos\left(\frac{2\pi k}{n}\right) = -1 \text{ if } n \text{ is odd}$$

Here, $n = 7$, and $k = 1, 2, 3, \dots, 6$, so sum is -1. If we sum over any 3 of these which are not symmetric partners, the sum is negative.

Alternatively, numerically, $\cos\left(\frac{2\pi}{7}\right) \approx 0.623$, $\cos\left(\frac{4\pi}{7}\right) \approx -0.222$, $\cos\left(\frac{6\pi}{7}\right) \approx -0.900$.

So $0.623 - 0.222 - 0.900 \approx -0.499$.

Conclusion: The sum is **negative**.

The reason: all three angles are in the first or second quadrant, but two of them produce negative cosines larger in magnitude than the positive one.

5. Suppose you start at $(1, 0)$ on the unit circle and travel a total distance of 8π in the clockwise direction. What point do you land on? Give your answer in coordinates and explain the pattern used.

Solution:

Circumference of unit circle is 2π . Clockwise corresponds to negative direction.

8π corresponds to 4 full revolutions ($8\pi/2\pi = 4$), so after 4 revolutions, you return to your starting point.

Therefore, you land at $(1, 0)$.

Pattern: Every full revolution returns to $(1, 0)$, so multiples of 2π , regardless of direction, return to the start.

6. Let t be a real number such that the terminal point on the unit circle is (x, y) . If t and $-t$ have the same reference number but different terminal points, describe fully the relationship between their coordinates, and give a nontrivial example.

Solution:

Reference number is the smallest positive angle to the x -axis. Both t and $-t$ have same reference number, but correspond to opposite directions around the circle.

The terminal points are $(x, y) = (\cos t, \sin t)$ and $(\cos(-t), \sin(-t)) = (\cos t, -\sin t)$.

So, the coordinates are related by reflection over the x -axis (negating the sine).

That is:

$$(x, y) \longleftrightarrow (x, -y)$$

Example: Take $t = \frac{2\pi}{3}$.

- $(\cos t, \sin t) = (\cos \frac{2\pi}{3}, \sin \frac{2\pi}{3}) = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$
- $(\cos(-t), \sin(-t)) = (\cos \frac{2\pi}{3}, -\sin \frac{2\pi}{3}) = (-\frac{1}{2}, -\frac{\sqrt{3}}{2})$

So, $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$ and $(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$ are the respective terminal points.
